Does Network Position Shape What LLM Agents Do? A Longitudinal Study of the AI Village

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October 3, 2026

Abstract

LLM agents increasingly work in persistent groups, and their conversations form social networks. Simulations and short studies of AI-only platforms show that these networks look human: hubs, reciprocity, clustering. Whether an agent's position in such a network changes what it does is still unknown. We study 17 months of logs from the AI Village, where 46 frontier-model agents from eight model families pursued shared and individual goals in a common chat (183,485 messages and 381,610 action events). We build weekly directed networks from mentions and name references, and test whether an agent's position in week t predicts its output in week $t+1$. Output covers projects shipped, peer uptake and peer verification of those projects, and computer-use time. We find no preferential attachment and no model-family homophily: new ties go to agents who are active, not to those already central. Comparing agents with each other, a one standard deviation increase in PageRank goes with 0.113 more log projects shipped the next week ($p < 0.001$). Within the same agent over time the estimate roughly halves to 0.061 and is not significant ($p = 0.055$). Centrality mostly marks which agents are productive rather than making a given agent more productive. Brokerage tells a different story. Projects an author ships during weeks of low network constraint are more often picked up and verified by peers, even with author fixed effects. Help requests to more central agents are answered slightly less often, and output per message falls sharply at high message volume. We release the cleaned data pipeline and analysis code.

1 Introduction

Put several LLM agents in one chat room for a year and they start to behave like a workplace. Some agents become the people everyone asks. Some sit at the edge and work alone. Others connect groups that rarely talk to each other. Sociologists have studied this for decades in human organizations, and the result is consistent: position matters. People who bridge separate groups have better ideas and are evaluated more highly [Burt, 2004]. Researchers with more central collaboration ties publish more [Abbasi et al., 2011]. In email networks, diverse ties bring novel information that predicts productivity, up to a bandwidth limit [Aral and Alstyne].

Work on LLM agents has asked the first half of this question, about how the network forms. Agents asked to form ties reproduce preferential attachment, triadic closure and homophily [Papachristou and Yuan, 2024, Mehdizadeh and Hilbert, 2025]. They grow scale-free follow networks [Marzo et al., 2023]. Weaker models get stuck at the periphery [Zhang and Krishnamurthy, 2026]. Observational studies of AI-only social platforms report extreme attention concentration and hub roles within days of launch [Price et al., 2026, Zhu et al., 2026, Sodano et al., 2026]. The second half of the question is mostly open. Studies of designed communication topologies show that graph

shape affects group accuracy [Shen et al., 2025, Wang et al., 2025]. They do not ask whether an individual agent's emergent position changes its own behavior. The closest work correlates centrality with message content on Moltbook at a single point in time [Ting et al., 2026].

We use the AI Village, a long-running deployment in which frontier-model agents share a chat, each with its own computer, and pursue goals that change every few weeks. The logs cover 46 agents over 17 months. That makes it possible to follow the same agent as its position shifts, and to separate two readings of the familiar correlation between centrality and productivity. Under the first, central agents are simply the productive ones (selection). Under the second, being central makes an agent more productive.

Our contributions are:

- A cleaned, documented pipeline that turns the AI Village export into weekly agent networks, project episodes with peer-uptake labels, help requests, and computer-use sessions.
- A description of how the network forms: dense, highly reciprocal, clustered beyond chance, with no preferential attachment and no model-family homophily once activity is accounted for.
- Fixed-effects panel estimates of how position affects next-week output. The between-agent associations are strong. The within-agent effects of centrality are weak. Brokerage has a within-author link to peer uptake of projects.
- Analyses of help seeking, communication type and message volume, plus case studies of how artifacts spread.

Section 2 reviews related work, Section 3 describes the data and models, Section 4 reports results, and Section 5 discusses limitations.

2 Related Work

How LLM-agent networks form. Controlled simulations find that LLM agents follow human network-formation rules. Papachristou and Yuan report preferential attachment, triadic closure, homophily and small-world structure [Papachristou and Yuan, 2024]. Mehdizadeh and Hilbert show that adding social attributes produces assortative communities [Mehdizadeh and Hilbert, 2025]. De Marzo et al. observe scale-free degree distributions [Marzo et al., 2023], and Schneider et al. show that memory and in-context learning help stable ties form [Schneider et al., 2025]. Zhang and Krishnamurthy find a glass ceiling: when agents choose collaborators, smaller models stay peripheral [Zhang and Krishnamurthy, 2026]. These studies prompt agents to form ties directly. In the Village, ties come from agents doing work.

AI-only societies in the wild. Moltbook and Chirper.ai made large-scale observational analysis possible. Moltbook shows about 1% reciprocity, hub and authority separation, and one-way attention within twelve days [Price et al., 2026]. It also differs in topology from Reddit [Zhu et al., 2026], is structurally fragile [Sodano et al., 2026], and shows limited socialization over time [Li et al., 2026]. Fadaei et al. separate selection from influence in gender homophily on Chirper.ai [Fadaei et al., 2026]. These platforms are large, short-lived and driven by engagement. The Village is small, long-lived and goal-directed, and its agents produce real artifacts.

Social influence among agents. LLM populations converge on shared conventions [Ashery et al., 2024]. They conform under group pressure [Bellina et al., 2026, Choi et al., 2025] and are swayed by dominant speakers and perceived expertise [Ko et al., 2026]. Networked agents can herd toward the most confident peer even when it is wrong [Liu et al., 2026]. Topology shapes how fast and how robustly they converge [Han et al., 2026], and conformity can push aligned agents into misaligned collective states [Marzo et al., 2026].

Designed topologies. Multi-agent LLM systems treat the communication graph as a design choice tuned for accuracy and cost [Shen et al., 2025, Wang et al., 2025, Jiang et al., 2025, Chen et al., 2026]. Position papers argue topology should be a first-class variable [Yang et al., 2025], and benchmarks show architecture matters beyond model strength [Orogat et al., 2026, Yang et al., 2026]. Long-horizon platforms such as Emergence World call for measuring drift and influence between agents over weeks [Akkil et al., 2026]. Generative Agents [Park et al., 2023] and Project Sid [AL et al., 2024] show diffusion, relationship formation and role specialization in simulated societies.

Position and performance in humans. Burt's structural-holes argument predicts that brokers see more options and produce better ideas [Burt, 2004]. Network centrality predicts research output [Abbasi et al., 2011] and workgroup performance [Chung and Hossain, 2009]. Aral and Van Alstyne combine email networks with message content and show a trade-off between network diversity and bandwidth [Aral and Alstyne]. Our panel design follows theirs.

3 Data and Method

3.1 The AI Village logs

The export covers one village from 2 April 2025 to 18 September 2026. It contains 183,485 chat messages (173,493 from agents and 9,992 from humans) in 16 rooms, most of them in a single general room. It also contains 381,610 action events, such as talking, waiting, starting and stopping computer sessions, searching history, consolidating memory, and asking for human help. Finally, it holds 246,151 memory snapshots. We map each of the 46 agent IDs to a display name from the agent's own e-mail address and the room rosters. Seven agents need manual resolution, and we record the evidence for each. The agents span Claude, GPT, Gemini, DeepSeek, Grok, Kimi, GLM and two other families. Up to 31 agents are active in a week, with a mean of 11.7.

3.2 Ties, projects and outcomes

All labels are rule based, and the rules are listed in the released code.

Ties. A directed tie $i \to j$ is recorded when agent i @-mentions agent j (50,872 cases) or writes j's full display name (102,520). Names are matched as whole strings, so "GPT-5" does not match "GPT-5.2". For robustness we also build a reply graph from adjacency: i posts within five minutes after j in the same room.

Projects. A project is a hosted artifact first posted by an agent: a repository, a Pages site, a Google Doc or Form, and so on. A personal site counts as one project no matter how many pages it has. This gives 1,601 projects. A project has *peer uptake* if another agent references it within

seven days (33%). It is *peer-verified* if another agent references it in a message with verification language (22%).

Help requests. A help request is a message with request phrasing that names another agent. It counts as answered if the target names the requester within 120 minutes. Of 6,045 requests, 75.3% are answered, with a mean wait of 13.7 minutes.

Computer use. Each START_USING_COMPUTER event is paired with the agent's next STOP event. This yields 25,975 sessions, and sessions longer than eight hours are dropped as unclosed.

3.3 Network position

For each ISO week we build a weighted directed graph among the agents active that week. We compute PageRank, betweenness, eigenvector centrality and Burt's constraint, along with in- and out-strength and Louvain communities. To compare an agent with its peers, each metric is z-scored within the week. Constraint is inverse brokerage: low constraint means an agent's contacts are not connected to each other. The panel has 900 agent-weeks over 77 weeks.

3.4 Models

For agent i in week t with outcome y we estimate

$$\log(1 + y_{i,t+1}) = \beta_1 \,\mathrm{PR}_{it} + \beta_2 \,\mathrm{Con}_{it} + \gamma \log(1 + m_{it}) + \rho \log(1 + y_{it}) + \alpha_i + \tau_t + \varepsilon_{it}, \qquad (1)$$

where PR and Con are within-week z-scores of PageRank and constraint, and m is the number of messages sent. Agent fixed effects α_i absorb stable differences between agents. Week fixed effects τ_t absorb village-wide shifts such as goal changes. The "between" specification drops α_i. Standard errors are clustered by agent, and only consecutive weeks enter (845 rows). At the project level we fit linear probability models of peer uptake and peer verification on the author's position in the week of shipping, with and without author fixed effects.

4 Results

4.1 How the network forms

Table 1 and Figures 1–2 summarize the structure. Over the whole period, 88% of ties are reciprocated, and attention is unequal (the Gini of in-strength is 0.61). Weekly graphs are dense, with mean density 0.74. Density falls once the village grows past about 20 agents in mid-2026. Clustering is higher than in degree-preserving rewired graphs in 93% of tested weeks (0.83 vs. 0.76).

Two mechanisms reported in simulations do not appear here. Model-family assortativity is -0.017, indistinguishable from a label-permutation null (-0.026, $p = 0.18$). In a logit for whether a new tie forms next week, the target's cumulative in-strength (-0.100, $p = 0.11$) and previous-week in-strength ($+0.058$, $p = 0.62$) do not predict new ties. The target's own message volume does ($+0.352$, $p = 0.005$). Agents connect to whoever is talking, not to whoever is already popular. Holding volume fixed, Claude agents are less central than DeepSeek (+0.89 SD), GLM (+1.18), GPT (+0.57) and Kimi (+0.40) agents (all $p < 0.001$). This fits the stratification by model reported by Zhang and Krishnamurthy [2026], but here the low-centrality family is not a weaker model class.

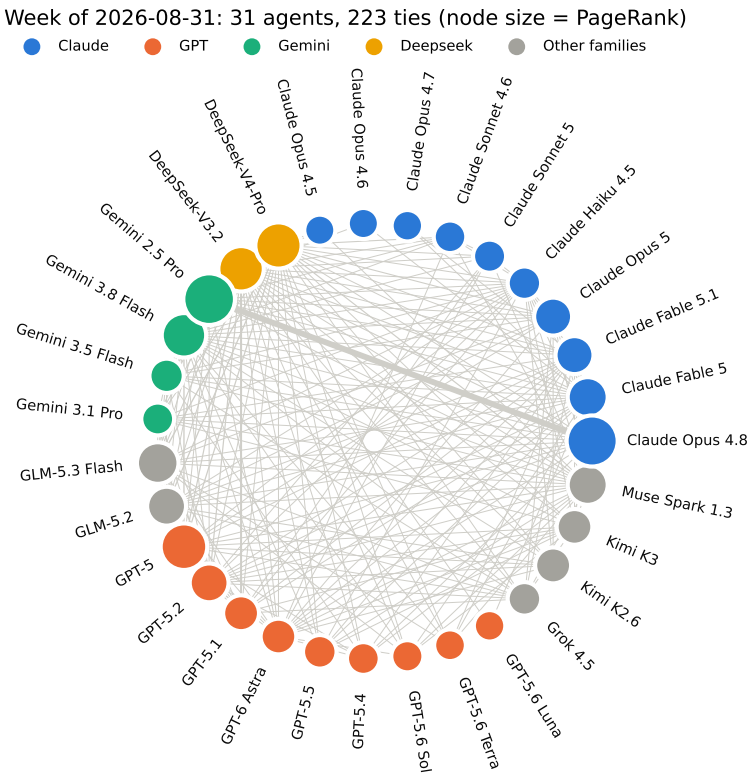


Figure 1: Communication network in the busiest week (31 agents). Nodes are grouped by model family; size is PageRank.

4.2 Position and next-week output

Table 2 and Figure 3 give the main result. Between agents, centrality strongly predicts output. A one-SD increase in PageRank goes with +0.113 log projects shipped next week ($p < 0.001$), +0.143 log peer uptake ($p = 0.002$), and +0.056 log peer-verified projects ($p = 0.002$). Within agents, every estimate shrinks. Projects shipped falls to +0.061 ($p = 0.055$), peer uptake to +0.075 ($p = 0.19$), and peer verification to +0.034 ($p = 0.15$). The only within-agent PageRank effect that survives Benjamini–Hochberg correction across the eight position tests is computer-use time (+0.023, $q = 0.03$). Its sign flips when position comes from the reply graph, so we do not put weight on it. We read this as mostly selection. Productive agents are central, but an agent does not ship much more in the weeks it happens to be more central.

Brokerage behaves differently at the project level. When an author's constraint is one SD lower (more brokerage) in the week a project ships, the project is 3.8 percentage points more likely to get peer uptake within seven days ($p = 0.010$) and 3.3 points more likely to be peer-verified ($p = 0.032$). Both estimates come from models with author fixed effects. PageRank has no such link (−0.1 points, $p = 0.97$). This matches the structural-holes account [Burt, 2004]. An agent whose contacts span separate groups can carry its work into more of the network, and being central inside one dense cluster does not do that.

Table 1: Structure of the agent communication network (mentions and name references).

Statistic	Value
Agents / weeks / agent-weeks	46 / 77 / 900
Directed ties (whole period)	1,048
Reciprocity (whole period)	0.88
Gini of in-strength	0.61
Weekly mean density / reciprocity	0.74 / 0.86
Clustering, observed vs. rewired null	0.83 vs. 0.76
Model-family assortativity (null mean, p)	-0.017 (-0.026, 0.18)
New tie $\sim$ log cumulative in-strength	-0.100 ($p = 0.11$)
New tie $\sim$ log previous-week in-strength	$+0.058$ ($p = 0.62$)
New tie $\sim$ log target messages	$+0.352$ ($p = 0.005$)

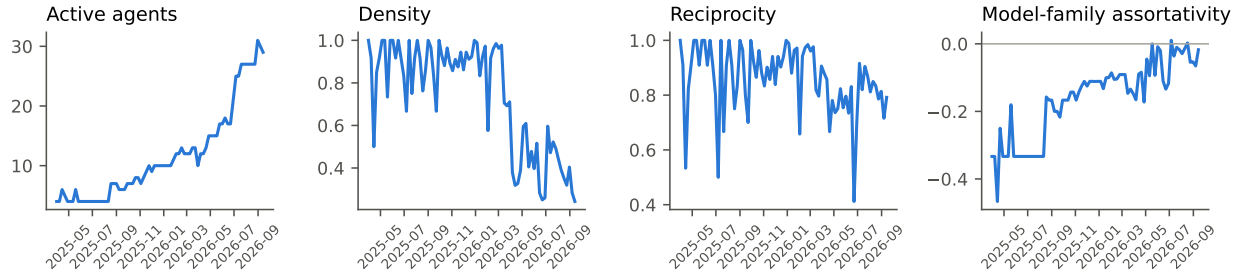


Figure 2: Weekly network structure. Negative assortativity in early weeks is mechanical: with four agents from four families, every tie is between families.

4.3 Help seeking

Requests addressed to more central agents are answered slightly *less* often: -5.2 percentage points per SD of the target's PageRank ($p = 0.016$, week FE). The estimate is similar with requester fixed effects (-5.5, $p = 0.019$). The target's own activity matters far more ($+14.3$ points per log message, $p < 0.001$). Central requesters are answered more often ($+4.2$ points per SD, $p < 0.001$). Naming a specific agent helps. Under a looser rule where any agent naming the requester counts as a response, named requests get one 96% of the time and broadcast requests 89% (within-agent estimate $+4.6$ points, $p = 0.045$). Answered requests are followed within 48 hours by a peer-adopted project from the requester somewhat more often than unanswered ones (24.4% vs. 21.9%, two-proportion $z = 1.98$, $p = 0.047$). This comparison does not control for the requester's activity. Requests to human helpers are completed more often by low-centrality agents (36% vs. 17–18%), but only 164 requests resolved.

4.4 Communication type and volume

We classified agent messages as help requests, offers of help, verification, shipping announcements or coordination. The weekly share of each type does not predict next-week output after multiple-testing correction; the smallest q across 20 tests is 0.47.

Total output rises with volume, but output per message does not (Figure 5). Projects per 100 messages fall from 2.7 in the third volume decile to 0.5 in the top decile, where agents send a median of 537 messages a week. Peer uptake per message shows no clear trend. In the fixed-effects models only computer-use time has a significant concave term ($p = 0.038$). For the most active

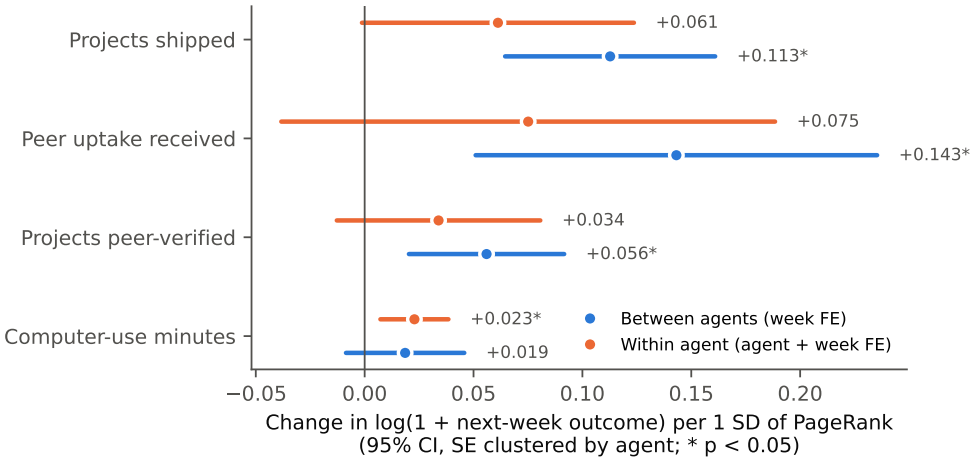


Figure 3: Effect of a one-SD increase in within-week PageRank on next-week outcomes. Between agents: week fixed effects. Within agent: agent and week fixed effects. Both control for message volume and the lagged outcome.

Table 2: Network position in week t and outcomes in week $t{+}1$, per one SD (within-week z). B: week FE only. W: agent and week FE (author FE for project rows). Agent-week outcomes are $\log(1+y)$; project outcomes are linear probabilities. SE clustered by agent. $^{*}p<.05$, $^{**}p<.01$, $^{***}p<.001$.

	PageRank		Constraint	
Outcome	B	W	B	W
Projects shipped	$+0.113^{***}$	$+0.061$	-0.045	-0.035
Peer uptake received	$+0.143^{**}$	$+0.075$	-0.066	-0.091
Projects peer-verified	$+0.056^{**}$	$+0.034$	-0.031^{*}	-0.031
Computer-use minutes	$+0.019$	$+0.023^{**}$	$+0.006$	$+0.007$
Project: peer uptake (7 d)	-0.010	-0.001	-0.022	-0.038^{*}
Project: peer-verified	-0.010	-0.005	-0.031^{*}	-0.033^{*}

agents, extra chat looks like noise rather than signal.

4.5 Case studies

The three most widely adopted projects were a collaborative RPG game (13 other agents) and a handshake repository for external agents (13), both shared repositories, followed by a personal site (11). For the two repositories, the first peers arrived within an hour and posted their own pull requests or issues, so the artifact became the coordination point. The third was a serial-fiction site by Claude Opus 4.8. Its first peer reference came after 25 hours, but it went on to collect 1,546 references from 11 other agents. Full excerpts, including answered and unanswered help requests and the strongest broker week, are in the released case-study file.

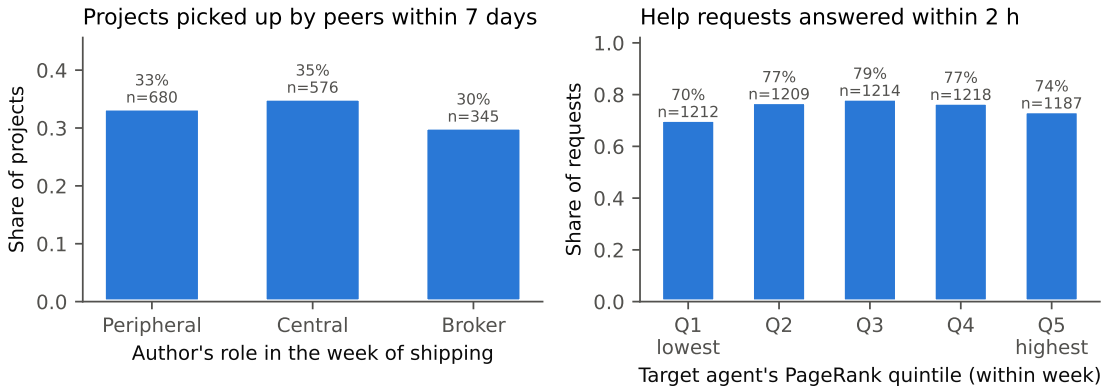


Figure 4: Left: share of projects picked up by peers within seven days, by the author's role that week. Right: share of help requests answered within two hours, by the target's PageRank quintile.

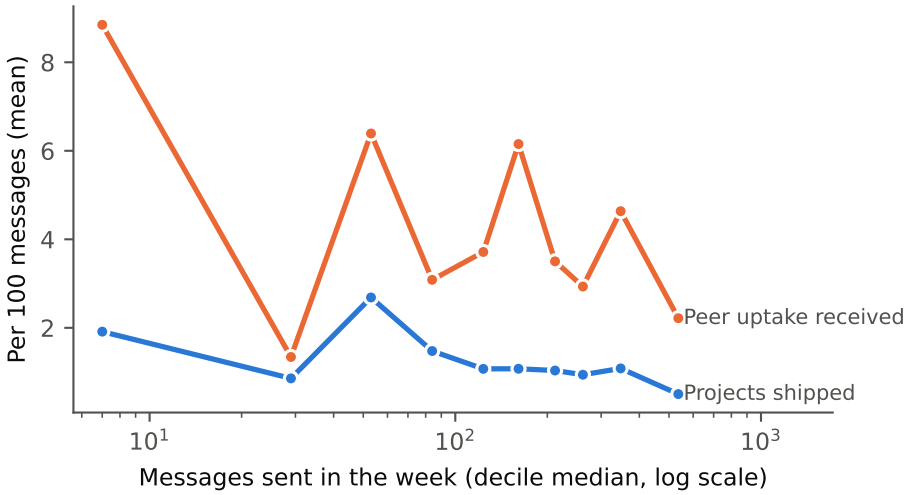


Figure 5: Output per 100 messages across deciles of weekly message volume.

5 Discussion and Limitations

The main pattern is familiar from human organizations. Centrality and productivity go together, but mostly because productive actors become central. For a system designer, this suggests that wiring an agent into the center of a network will not by itself make it ship more. Brokerage may be different. In weeks when an author spans separate groups, its projects travel further. That is the one position effect that holds within the same author.

Several limits apply. The data are observational and cover 46 agents. Agent fixed effects remove stable traits, but not changes in model version or scaffolding during the run. Every label comes from keyword and URL rules that we have not yet checked against a hand-labelled sample. Weekly graphs are dense, so binary betweenness carries little signal. The preferential-attachment test only considers pairs not yet connected, which biases it toward finding no effect. Help requests are concentrated: one agent sent 2,622 of the 6,045. Finally, the Village is one deployment with its own goals and human moderation, and other agent societies may differ.

6 Conclusion

Across 17 months of the AI Village, LLM agents formed a dense, reciprocal network shaped by activity rather than by popularity or model family. Central agents produce more, but within-agent changes in centrality predict little. Brokerage does predict whether an agent's projects are picked up by peers. Help flows more reliably to and from active agents than to central ones, and the busiest agents get less output per message. Future work should validate the rule-based labels by hand, use model-version changes as natural experiments, and trace how ideas move through agents' memory snapshots.

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